

PLAN DE TEST

Realizat de: Lapuste Andrei Raul

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1. Introducere

“HRHelper” este o aplicație concepută propriu de-al lungul semestrului, concepută după tema unei aplicații web folosită între angajații unei firme pentru a comunica unul cu altul, de a-și managera propria listă de clienți și a trimite emailuri între utilizator și client.

Întreaga aplicație presupune 3 pagini web, pentru logarea utilizatorului la server și a dobândi acces la restul funcționalităților, pagina principală folosită pentru a vizualiza lista clienților, informații despre fiecare client în parte, adăugarea unui nou client, respectiv ștergerea acestuia și trimiterea mesajelor. A treia pagina este folosită pentru vizualizarea mesajelor transmise între utilizatori și primirea mesaje automate din partea serverului. Partea back-end a aplicației prevede un server propriu format din 6 microservicii individuale. Serviciul de memorie, ce scop are de a urmări statusul online, respectiv online al utilizatorului, serviciul de logare, serviciul de manipulare al mesajelor, de manipulare al clienților, un serviciu separat, pentru transmiterea mesajelor printr-o adresă de email prin internet, destinate unui client, și un serviciu adminin, utilizat de administratorul desemnat aplicației pentru a urmării utilizatorii, de a revoca accesul la aplicație și de a oferi posibilitatea procesului de schimbare al parolei printr-un link trimis la o adresă de email aleasă personal de el/ea. Baza de date este una comună, formată din trei tabele pentru stocarea mesajelor, stocarea clienților și stocarea informațiilor despre fiecare utilizator.

2. Scopul și obiective

Actualul plan de testare presupune de a investiga funcționalitățile aplicației web din punctul de vedere al utilizatorilor și administratorilor:

- Logarea la aplicație
- Utilizarea resurselor destinate managementului clienților
- Comunicarea cu utilizatori separați și clienți
- Vizualizarea mesajelor
- Rezultatele în urma acțiunilor administrative

3. Strategii de testare

Aplicația va fi testată pentru toate funcționalitățile puse la dispoziția utilizatorilor și administratorului

3.1 Testarea funcțională

Acest tip de testare este folosit pentru verificarea funcționalităților unei aplicații deja în starea de a fi utilizată într-o oarecare măsură de testări la nivelul de utilizator.

Testarea funcționalităților paginii web pentru logare se va testa manual:

- Logarea în aplicație folosind email și parolă valide- rezultă deschiderea paginii principale
- Logarea în aplicație folosind email sau parolă invalidă- apariția unui mesaj eroare
- Logarea în aplicație când serverul e picat- apariția unui mesaj de conexiune
- Logarea în aplicație când utilizatorul este deja logat- apariția unui mesaj pentru acces neautorizat

Testarea funcționalităților paginii principale se va face manual:

- Trimiterea unui mesaj după selectarea unui contact- apariția unui răspuns adecvat situației
- Selectarea unui contact- afisarea informațiilor acestuia
- Ștergerea contactului după selectarea acestuia și apăsarea butonului de ștergere- dispariția acestuia din listă
- Completarea formularului de adăugare al contactului- apariția acestuia în lista de contacte
- Deconectarea de la aplicație- întoarcerea la pagina de logare

Testarea funcționalităților paginii de recepție a mesajelor:

- Vizualizarea listei mesajelor la deschiderea paginii
- Dublu-click mesaj- expansiunea acestuia

3.2 Testarea de integrare

Acest tip de testare presupune analizarea subfuncțiilor aplicației neaccesibile direct de către utilizator, și a căror interacțiuni cu serverul sau aplicația însuși trebuie testate și urmărite în timp real.

Testarea procedurilor interne ale paginii de logare:

- Logarea corectă- stocarea unui tuplet de tip șir JSON în sessionStorage ce cuprinde informațiile utilizatorului

Testarea procedurilor interne ale paginii principale:

- Accesarea paginii în urma logării - accesul la funcționalitățile aplicației "HRHelper"
- Accesarea paginii fără a se loga legal- mesaj de eroare, lipsa funcționalităților
- Dacă accesul a fost revocat- deconectarea forțată din aplicație
- Picarea serverului- mesaj cauzat de lipsa de conexiune

- Apariția listei de contacte a utilizatorului- clienți și alti utilizatori în partea stângă a ecranului în urma logării reușite.

Testarea procedurilor interne ale paginii de recepție a mesajelor

- Încărcarea listei mesajelor la deschiderea paginii

4. Componente care vor fi testate

1. Funcționalitățile paginii de logare
2. Funcționalitățile paginii principale
3. Funcționalitățile paginii de mesaje
4. Stabilitatea conexiunii interne cu serverul
5. Schimbarea parolei prin serviciul admin
6. Revocarea accesului prin serviciul admin

5. Componente care nu vor fi testate

- Servicile serverului în mod direct prin instrumente de testare

6. Participanți

1. Lăpuște Andrei Raul

7. Resurse

7.1 Resurse hardware

- Calculator
- Imprimantă

7.2 Resurse software

- Windows
- Visual Studio Code
- Microsoft Visual Studio
- Swagger

8. Anexa

8.1 Test Case 1

Test Case 8.1	Test Case Name: Success_login	Page 1 of 1
System: HRHelper_Login	Subsystem: Logging microserver	
Designed by: Lapuste Raul	Design Date: 09/06/2025	
Executed by: Lapuste Raul	Execution Date: 09/06/2025	
Short Description: Test logging with valid user email and password		

Pre-conditions

1. The user has a valid email and password
 2. User is offline and access is not restricted
- Current email of the user: 'arlapuste@ttp.com'
- Current password of the user: 'Popoye'

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Click the 'log.html' file	'HRHelper Login' will open in browser	Pass	
2	Enter valid email		Pass	Not expecting response
3	Enter valid password		Pass	Not expecting response
4	Press 'Login' button	No error message from microserver	Pass	Access granted
5	Check Post-condition 1		Pass	

Post-conditions

1. Redirected to 'HRHelper_Main' page

8.2 Test Case 2

Test Case 8.2	Test Case Name: Missing_Email_Password	Page 1 of 1
System: HRHelper_Login	Subsystem: Logging microserver	
Designed by: Lapuste Raul	Design Date: 09/06/2025	
Executed by: Lapuste Raul	Execution Date: 09/06/2025	
Short Description: Test logging function when no email or password is used		

Pre-conditions

- 1.The user has a valid email and password
 - 2.User is offline and access is not restricted
- Current email of the user: 'arlapuste@ttp.com'
- Current password of the user:'Popoye'

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Click the 'log.html' file	'HRHelper_Login' will open in browser	Pass	
2	Press 'Login' button	Not redirected to 'HRHelper' main page	Pass	Still in 'HRHelper_Login'
3	Check Post-condition 1		Pass	
4	Check Post-condition 3		Pass	
5	Enter valid email		Pass	Not expecting response
6	Press 'Login' button	Not redirected to 'HRHelper' main page	Pass	Still in 'HRHelper_Login'
7	Check Post-condition 2		Pass	
8	Enter valid password		Pass	Not expecting response
9	Press 'Login' button	Not redirected to 'HRHelper' main page	Pass	Still in 'HRHelper_Login'
10	Check Post-condition 4		Pass	
11	Check Post-condition 5		Pass	

Post-conditions

1. Red message 'User is required' will show up under email textbox
2. Red message 'User is required' disappears
3. Red message 'Password is required' will show up under password textbox
4. Red message 'Password is required' disappears
5. Redirected to 'HRHelper' main page

8.3 Test Case 3

Test Case 8.3

Test Case Name: [Email/Password Wrong](#) Page 1 of 1

System: [HRHelper_Login](#)

Subsystem: [Logging microserver](#)

Designed by: [Lapuste Raul](#)

Design Date: [09/06/2025](#)

Executed by: [Lapuste Raul](#)

Execution Date: [09/06/2025](#)

Short Description: [Test logging function when email or password is wrong](#)

Pre-conditions

- 1.The user has a valid email and password
 - 2.User is offline and access is not restricted
- Valid email of the user: '[arlapuste@ttp.com](#)'
Invalid email of the user: '[ar222lapuste33@ttp.com](#)'
Valid password of the user: 'Popoye'
Invalid password of the user: 'Bluto'

Step	Action	Expected System Response	Pass/Fail	Comment
1	Click the 'log.html' file	'HRHelper_Login' will open in browser	Pass	
2	Enter valid email		Pass	Not expecting response
3	Enter invalid password		Pass	Not expecting response
4	Press 'Login' button	Not redirected to 'HRHelper' main page	Pass	Still in 'HRHelper_Login'
5	Check Post-condition 1		Pass	
6	Enter invalid email		Pass	Not expecting response
7	Enter valid password		Pass	Not expecting response
8	Press 'Login' button	Not redirected to 'HRHelper' main page	Pass	Still in 'HRHelper_Login'
9	Check Post-condition 1		Pass	

Post-conditions

1. Alert showing microserver message 'EMAIL OR PASSWORD WRONG!'

8.4 Test Case 4

Test Case 8.4

Test Case Name: [Logging when restricted](#) Page 1 of 1

System: [HRHelper_Login](#)

Subsystem: [Logging microserver](#)

Designed by: [Lapuste Raul](#)

Design Date: [09/06/2025](#)

Executed by: [Lapuste Raul](#)

Execution Date: [09/06/2025](#)

Short Description: [Test logging function when user access is restricted](#)

Pre-conditions

1.The user has a valid email and password

2.User is offline

3.Access is restricted by admin

Valid email of the user: '[arlapuste@ttp.com](#)'

Valid password of the user: 'Popoye'

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Click the 'log.html' file	'HRHelper_Login' will open in browser	Pass	
2	Enter valid email		Pass	Not expecting response
3	Enter valid password		Pass	Not expecting response
4	Press 'Login' button	Not redirected to 'HRHelper' main page	Pass	Still in 'HRHelper Login'
5	Check Post-condition 1		Pass	

Post-conditions

1. Alert showing microserver message 'YOU ARE BLOCKED!'

8.5 Test Case 5

Test Case 8.5

Test Case Name: [Server down logging](#)

Page 1 of 1

System: ['HRHelper_Login'](#)

Subsystem:

Designed by: [Lapuste Raul](#)

Design Date: [09/06/2025](#)

Executed by: [Lapuste Raul](#)

Execution Date: [09/06/2025](#)

Short Description: [Test logging page procedure when server is down](#)

Pre-conditions

- 1.The user has a valid email and password
- 2.User is offline

Server is offline

Valid email of the user: ['arlapuste@ttp.com'](#)

Valid password of the user: ['Popoye'](#)

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Click the 'log.html' file	'HRHelper_Login' will open in browser	Pass	
2	Enter valid email		Pass	Not expecting response
3	Enter valid password		Pass	Not expecting response
4	Press 'Login' button	Not redirected to 'HRHelper' main page	Pass	Still in 'HRHelper_Login'
5	Check Post-condition 1		Pass	

Post-conditions

1. Alert showing microserver message ['NU AM PUTUT STABILI CONEXIUNEA!'](#)

8.6 Test Case 6

Test Case 8.6

Test Case Name:

Delogging and Logging back

Page 1 of 1

System: HRHelper_Login & HRHelper_Main

Subsystem: Logging microserver

Designed by: Lapuste Raul

Design Date: 09/06/2025

Executed by: Lapuste Raul

Execution Date: 09/06/2025

Short Description: Testing delogging and logging back in the web application

Pre-conditions

- 1.The user has a valid email and password
 - 2.User 'arlapuste@ttp.com' already is online
- Valid email of the user: 'arlapuste@ttp.com'
Valid password of the user: 'Popoye'

Step	Action	Expected System Response	Pass/Fail	Comment
1	Check Pre-condition 2		Pass	
2	Press 'LOG OUT' button	Message of ' GOOD BYE'	Pass	
3	Check Post-condition 2		Pass	
4	Enter valid email		Pass	Not expecting response
5	Enter valid password		Pass	Not expecting response
6	Press 'Login' button	No error message from microserver	Pass	Access granted
7	Check Post-condition 1		Pass	

Post-conditions

1. User is in 'HRHelper' main page
2. Redirected to 'HRHelper_Login'

8.7 Test Case 7

Test Case 8.7

Test Case Name: [Connection with server](#) Page 1 of 1

System: [HRHelper_Main](#)

Subsystem: [Memory microservice](#)

Designed by: [Lapuste Raul](#)

Design Date: [09/06/2025](#)

Executed by: [Lapuste Raul](#)

Execution Date: [09/06/2025](#)

Short Description: [Testing communication procedure with the server](#)

Pre-conditions

- 1.User 'arlapuste@ttp.com' already is online
- 2.User email was saved in sessionStorage as 'jk'

Step	Action	Expected System Response	Pass/Fail	Comment
1	Check Pre-condition 1		Pass	
2	Press 'F12'	Open browser console	Pass	
3	Await 15 seconds	Message 'You are still connected'	Pass	
4	Enter command in Console: 'sessionStorage.getItem('jk')'	JSON string of user's email and pass: "granted"	Pass	
5	Check Post-condition 1		Pass	

Post-conditions

- 1.Contact list loaded with page

8.8 Test Case 8

Test Case 8.8

Test Case Name: Network messaging

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System: HRHelper_Main

Subsystem: Message microserver

Designed by: Lapuste Raul

Design Date: 09/06/2025

Executed by: Lapuste Raul

Execution Date: 09/06/2025

Short Description: Testing sending message
function to a coworker

Pre-conditions

1. User 'arlapuste@ttp.com' already is online
 2. List of contacts is not empty
 3. Selected contact's email ends with domain '@ttp.com'
- Text message : 'Time=12:43'

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Press 'Send' button	System shows 'PLEASE WRITE A MESSAGE'	Pass	
2	Insert message in textbox		Pass	Not expecting response
3	Press 'Send' button	System shows ' PLEASE SELECT FROM LIST'	Pass	
4	Double-click contact	System shows contact information and readies for sending message	Pass	
5	Press 'Send' button	System shows message ' MESSAGE DELIVERED'	Pass	
6	Check Post-condition 1		Pass	Checked using valid user account

Post-conditions

1. Message saved and sent to coworker contact account

8.9 Test Case 9

Test Case 8.9

Test Case Name: Delete Contact

Page 1 of 1

System: HRHelper_Main

Subsystem: Person microserver

Designed by: Lapuste Raul

Design Date: 10/06/2025

Executed by: Lapuste Raul

Execution Date: 10/06/2025

Short Description: Testing function to delete a contact from list

Pre-conditions

1. User 'arlapuste@ttp.com' already is online
2. List of contacts is not empty

Step	Action	Expected System Response	Pass/Fail	Comment
1	Press 'Delete Contact' button	System shows ' SELECT SOMEONE ALREADY IN THE LIST'	Pass	
2	Double-click contact	System shows contact information and readies it for deletion	Pass	
3	Press 'Delete Contact' button	System shows ' PERSOANA STEARSA'	Pass	
4	Reload the page 'HRHelper Main'	System reload without problems	Pain	
5	Check Post-condition 1		Pass	

Post-conditions

1. Contact was deleted completely from the list

8.10 Test Case 10

Test Case [8.10](#)

Test Case Name: [Add Contact](#)

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System: [HRHelper_Main](#)

Subsystem: [Person microserver](#)

Designed by: [Lapuste Raul](#)

Design Date: [10/06/2025](#)

Executed by: [Lapuste Raul](#)

Execution Date: [10/06/2025](#)

Short Description: [Testing function to add a new contact](#)

Pre-conditions

1. User 'arlapuste@ttp.com' already is online

Fields to complete:

1. Nume: Andrei Raul
2. Firma: Ttp SRL
3. Nr Telefon: 22 22 22 22
4. Email: raullapustel@gmail.com
5. Info(Optional):

Step	Action	Expected System Response	Pass /Fail	Comment
1	Press 'ADD CONTACT ' button	System will open a form for completing fields	Pass	
2	Complete the specified fields		Pass	Not expecting response
3	Press 'Add Contact'button	System will automatically reload the page	Pass	
4	Check Post-condition 1		Pass	

Post-conditions

1. Contact was added successfully

8.11 Test Case 11

Test Case 8.11	Test Case Name: Add Contact Faulty	Page 1 of 1
System: HRHelper_Main	Subsystem: Person microserver	
Designed by: Lapuste Raul	Design Date: 10/06/2025	
Executed by: Lapuste Raul	Execution Date: 10/06/2025	
Short Description: Testing function to add a new contact without email		

Pre-conditions

1. User 'arlapuste@tpp.com' already is online

Fields to complete:

6. Nume: Ana Georgiana
7. Firma: Rolex Uk
8. Nr Telefon: 09 09 09 09
9. Email:
10. Info(Optional):

Step	Action	Expected System Response	Pass /Fail	Comment
1	Press 'ADD CONTACT ' button	System will open a form for completing fields	Pass	
2	Complete the specified fields		Pass	Not expecting response
3	Press 'Add Contact'button	System will show error message "Can't add person"	Fail	
4	Check Post-condition 1		Fail	

Post-conditions

1. Contact was not added

8.12 Test Case 12

Test Case 8.12	Test Case Name: View Emails	Page 1 of 1
System: HRHelper_Email	Subsystem: Message microserver	
Designed by: Lapuste Raul	Design Date: 10/06/2025	
Executed by: Lapuste Raul	Execution Date: 10/06/2025	
Short Description: Testing procedures of HRHelpers Email message page		

Pre-conditions

1. User 'arlapuste@ttp.com' already is online

Step	Action	Expected System Response	Pass/Fail	Comment
1	Press 'EMAILS' option from navigation bar	System will open 'HRHelper_Email' page	Pass	
2	Check Post-condition 1		Pass	
3	Check Post-condition 2		Pass	
4	Double-click the message	System will extend the message body	Pass	
5	Repeat step 4	System will collapse the message body	Pass	
6	Press 'EMPLOYEES' option of the navigation bar	System will open 'HRHelper_Main' page	Pass	
7	Check Post-condition 3		Pass	

Post-conditions

1. Rerouted to 'HRHelper_Email'
2. Email list is loaded with the page
3. Rerouted to 'HRHelper_Main'

8.13 Test Case 13

Test Case 8.13	Test Case Name: Outside Mail	Page 1 of 1
System: HRHelper_Main	Subsystem: Outside_Email microserver	
Designed by: Lapuste Raul	Design Date: 10/06/2025	
Executed by: Lapuste Raul	Execution Date: 10/06/2025	
Short Description: Testing procedure to send an message over internet via email		

Pre-conditions

1. User 'arlapuste@ttp.com' is already online
2. Selected contact has a valid Gmail address: 'raullapuste1@gmail.com'
3. Text message : 'Time=15:03'

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Press 'Send' button	System shows 'PLEASE WRITE A MESSAGE'	Pass	
2	Insert text message in textbox		Pass	Not expecting response
3	Press 'Send' button	System shows ' PLEASE SELECT FROM LIST'	Pass	
4	Double-click user	System shows contact information and readies for sending message	Pass	
5	Check Pre-condition 2		Pass	
5	Press 'Send' button	System shows message ' MESSAGE DELIVERED'	Pass	
6	Press 'EMAILS' option from navigation bar	System will open 'HRHelper_Email' page	Pass	
7	Check Post-condition 1		Pass	
8	Check Post-condition 2		Pass	
9	Check Post-condition 3		Pass	

Post-conditions

1. Page 'HRHelper_Email' was loaded
2. Email list is loaded with the page
3. No failure message was sent to user

8.14 Test Case 14

Test Case 8.14	Test Case Name: Outside Mail	Page 1 of 1
System: ADMIN FORM	Subsystem: Admin microserver	
Designed by: Lapuste Raul	Design Date: 10/06/2025	
Executed by: Lapuste Raul	Execution Date: 10/06/2025	
Short Description: Testing function offered by server to change user password		

Pre-conditions

1. User 'arlapuste@ttp.com' asks to change password
2. User provides a valid email address: 'raullapuste1@gmail.com' to the admin
3. User personal password ''*****''
4. Body of method: 'raullapuste1@gmail.com_arlapuste@ttp.com'
5. New password to be choose: 'Bluto12'

Step	Action	Expected System Response	Pass /Fail	Comment
1	Admin calls method 'Send Link Password Changer'	System returns status 200 for sent email	Pass	
2	User connects to personal account with provided address and personal password	System provides access to account	Pass	
3	Check Post-condition 1		Pass	
4	Check Post-condition 2	System shows contact information and readies for sending message	Pass	
5	User clicks email	System opens email	Pass	
5	Check Post-condition 3		Pass	
6	User clicks https link	System will open 'ADMIN FORM' page	Pass	
7	Check Post-condition 4		Pass	
8	User enters new password for both texboxes		Pass	Not expecting response
9	Press 'CHANGE PASSWORD'	System will response with message from server	Pass	
10	Check Post-condition 5		Pass	
11	Execut Test Case 8.1 for password 'Bluto12'		Pass	

Post-conditions

1. Account lists the inbox emails
2. Email 'no_reply_email' exists in inbox
3. Email contains 'Please choose your password via link below' followed by a https link
4. 'ADMIN FORM' page was loaded successfully
5. System displays 'PASSWORD WAS UPDATED', page 'ADMIN FORM' becomes not found